Chapter 1: Coupled Simulation

Now we couple locally mass conserved MFEM and FVM method together in this chapter and perform coupled numerical simulation for partially molten mantle. We discuss implementation details and coupling strategy combining MFEM, FVM and eutectic phase package. The remaining uncertainty of porosity ϕ will be fulfilled by phase package evaluation. Before we throw ourselves into the fully coupled problem. We test another porosity evolution scenario with a more straightforward prescribed melting. Then we will show an example combining MFEM, FVM and eutectic phase package altogether and perform coupled computation. We start with pseudo 1D scenarios and then proceed to perform full 2D simulation.

1.1 Coupled Algorithm

Let $\mathcal{U}^n = (C^n, H^n)$ be the cell-averaged solution to the transport system (**??**) at time t^n. Let $\mathcal{V}^n = (\check{\mathbf{v}}_r^n, \check{\mathbf{v}}_s^n, \check{q}_l^n, \check{q}_s^n)$ denote the solution to the Darcy-Stokes system at the same time step t^n. Moreover, let $\mathcal{W}^n = (\phi^n, \phi_1^n, \check{T}^n, c_s^n, c_l^n)$ denote the output of the phase calculation at t^n given the transport result U^n.

For each time step t^n, we reconstruct dimensionless mass composition and enthalpy from the cell-averaged solution $\mathcal{U}^n$ with ML-WENO reconstruction at each data points. The eutectic phase package gives corresponding phase parameters $\mathcal{W}^n$ according to these point wise values. Once we know $\mathcal{W}^n$, we solve a fixed in time Darcy-Stokes system at this time step t^n for $\mathcal{V}^n$. The cell-averaged solution $\mathcal{U}^{n+1}$ of the next time step t^{n+1} is then obtained by forward marching the system explicitly. We formally formulate an overall explicit coupled solver as shown in algorithm 1.

Algorithm 1 Sequential Coupled Solver

Let U^0 be given as initial data.

1: Initialize $\mathcal{U}^0$, compute $\mathcal{W}^0$ with eutectic phase package.

2: Compute ϕ^0 in $\mathcal{W}^0$, solve the Darcy-Stokes system (**??**)–(**??**) for $\mathcal{V}^0$.

3: **for** Time level $n = 1, 2, \ldots, n_{\max} - 1$ **do**

4: Given $\mathcal{U}^n$, compute $\mathcal{W}^n$ by the eutectic phase package.

5: Compute ϕ^n in W^n, solve the Darcy-Stokes system (**??**)–(**??**) for $\mathcal{V}^{n+1}$.

6: Given $\mathcal{W}^n$ and $\mathcal{V}^{n+1}$, compute dimensionless effective velocity (**??**) and phase averaged velocity as

$$\check{\mathbf{v}}_{\text{eff}} = \frac{\phi \check{\mathbf{v}}_l + D_i(1-\phi)\check{\mathbf{v}}_s}{\phi + (1-\phi)D_i}, \quad \check{\bar{\mathbf{v}}} = \phi \check{\mathbf{v}}_r + \check{\mathbf{v}}_s.$$

7: Explicitly marching forward the transport system (**??**) to next time $n+1$ for $\mathcal{U}^{n+1}$.

8: **end for**

1.1.1 Discontinuous Porosity

Due to the presence of no melt regions and different phase regions, we may obtain different reconstructed values of dimensionless enthalpy and mass concentration from two sides of an edge shared by two elements and thus two different calculated porosities. However, it is crucial to have one compatible and appropriate porosity value used on the element edge in the conforming MFEM scheme.

Therefore, we evaluate the geometric average of the two values of porosity ϕ_e^+ and ϕ_e^- at each quadrature points on both sides of an element edge e as

$$\overline{\phi}_e = \frac{2\phi_e^+\phi_e^-}{\phi_e^+ + \phi_e^-}, \tag{1.1}$$

where ϕ_e^+ and ϕ_e^- are porosity evaluated from two sides of the element edge e. The geometric averaged porosity $\overline{\phi}_e$ is required for the continuous compaction equation in Darcy equation (**??**) , where face integral is replaced with edge integral through divergence theorem. When the porosity on one side equal zero, the geometric average of the porosity equals zero as well, which highlights the fact that there is no liquid flow should travel through this edge.

1.2 Pseudo One-Dimensional Column Tests

In this section, we first test our coupled algorithm on a 2D $M \times N$ grid with a pseudo-1D column problem. We test two different scenarios, one couples MFEM and FVM solver without phase computation but with a simple prescribed melting function $\Gamma(\mathbf{x})$ instead. The other one exhibits a fully coupled scenario where phase transition and porosity evolution present.

For the two tests presented below, we fix mesh resolution as $M = 2$ and $N = 100$. Following the

1.2.1 Prescribed Melting Case

Before fully coupled computation, we illustrate a simpler model coupling case coupling MFEM and FVM solver. Instead of deciding porosity with phase package computation, we prescribe the melting rate or the change of porosity as a fixed function distributed in space as

$$\phi(t) = \Gamma(\mathbf{x}). \tag{1.2}$$

We write conservation of liquid mass with the prescribed melting term Γ (1.2) as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_l) = \Gamma, \tag{1.3}$$

where diffusion effect is being dropped for simplicity in this case.

We prescribe the following boundary conditions for MFEM and FVM solver respectively. For the Darcy-Stokes MFEM solver:

- Bottom Edge $y = 0$ and $y = -0.4$;

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions:

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0,\ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu}\, ds = 0.$$

- Tangential Stokes velocity is prescribed with essential boundary condition:

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0.$$

- Normal Darcy flux is prescribed with essential boundary condition:

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e u\, ds = 0.$$

- Top Edge $y = 0$;

- Left and Right Edges $x = \pm L$;

  - Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions:

  $$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0,\ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu}\, ds = 0;$$

  - Tangential Stokes velocity is prescribed with the natural boundary condition: free traction

  $$t_0 = 0;$$

  - Normal Darcy flux is prescribed with essential boundary condition:

  $$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e u\, ds = 0.$$

For FVM solver:

- Bottom boundary;

For simplicity and efficiency in computation, we use 1×3 and 1×2 for our ML-WENO(3,2) stencils, since we don't need resolution and high order accuracy in x direction. We show two different melting source scenarios. One of which is a piece wise constant stair function:

$$\Gamma_1(z) = \begin{cases} 1e-3, & -0.2 < z \leq 0, \\ 5e-4, & -0.3 < z \leq -0.2, \\ 0, & -0.4 \leq z \leq -0.3, \end{cases} \tag{1.4}$$

and one with melting source trapped in the middle of the melting column as

$$\Gamma_2(z) = \begin{cases} 0, & -0.2 < z \leq 0, \\ 5e-4, & -0.3 < z \leq -0.2, \\ 0, & -0.4 \leq z \leq -0.3, \end{cases} \tag{1.5}$$

We assign different initial conditions to these two scenarios as

$$\phi_1(z, 0) = 0.0, \quad z \in [-0.4, 0.0], \tag{1.6}$$

and

$$\phi_2(z, 0) = \begin{cases} 1e-2, & -0.2 < z \leq 0, \\ 0, & -0.4 \leq z \leq -0.2. \end{cases} \tag{1.7}$$

1.2.2 Phase Transition Case

1.3 Two-Dimensional Test

1.4 Parallel Implementation

In this section, we disscuss some ideas regarding parallel implementation of MFEM and FVM solvers, and present some preliminary scability results for each solver. The parallel implementation of the MFEM and FVM need different treatments.

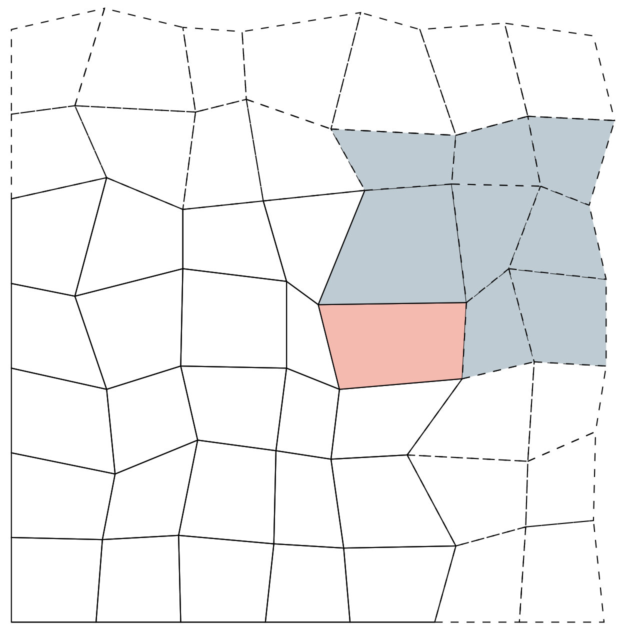

Figure 1.1: An exhibition of ghost region in parallel implementation of ML-WENO scheme.

For FVM discretization, the high order stencils posed some difficiulties on parallel implementation. We need to extend from one processor to the adjecent processor with the so-called ghost region, which results in more communication during computation. Consider a ML-WENO(3,2) scheme, as shown in Figure 1.1, we show distributed ghost region with dashed lines. We need to determine the flux acorss the edge of the red cell. Therefore, a maximum 3×3 stencil should be attached to the outside cell. Hence, in a ML-WENO(3,2) scheme, a two-cell ghost layer should be distributed and updated everytime computing flux.

For MFEM discretization, we encounter different problems. The MFEM discretization, unlike FVM scheme, forms a sparse matrix. The sparse matrix are splited across the processors. The dof in solution vector should be distributed accordingly. Therefore, everytime the solution is being reconstructed, communication between processors will be induced.

We provide strong scaling plot for our test parallel running. We run our MFEM and FVM code with 1,2,4,6 processors

1.4.1 Parallel Index Convention

Index convention in parallel computing requires another "local" index kept within each processor. We can bypass such difficultity by using Hash table data structure that directly mapping to global index with out adding another "local" index. However, in the worst case Hash table suffers $O(n)$ time complexity in searching.

1.5 Data Management with PETSc

PETSc, the Protable, Extensible Toolkit for Scientific Computation has been gaining popularity in scientific applications modelled by partial differential equations **?** and **?**. We used three main features from PETSc package.

- Vector (Vec) and matrix (Mat) data structure.
- Data management object (DM).
- Standard linear solver (KSP).

We store global data in Vec and Mat data structure provided by PETSc. Vec and Mat objects in PETSc can complete simple add, dot operations without distinguish between sequential or parallel.

Note that PETSc provides a built-in data management object DMStag. However, we use a more generic data management object DMDA for structured mesh because we formed reduced linear system in MFEM computation. DMStag, on the other hand, does not provide a straightforward function that can create reduced system for us.

We use the built in KSP solver, e.g., KSPMINRES and KSPCG solver for our application when we solve large sparse system. However, for small system such as 4×4 system solved in ML-WENO stencil polynomial computation, we call lapack directly. We also use provided preconditioner.

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